

SPETC: An Open-Source Spectro-Photometric Exposure Time Calculator for Small Telescopes and Amateur Instrumentation

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Abstract

SPETC is an open-source, offline exposure time calculator for ground-based broad-band photometry and slit or slitless spectroscopy, aimed at the instrumentation actually used on small telescopes: CMOS cameras with gain-dependent read noise, one-shot-colour sensors, classical slit spectrographs, and Star Analyser transmission gratings. It couples CALSPEC-calibrated template spectra and SVO-convention synthetic photometry to a position- and time-dependent sky model (airglow, zodiacal light, starlight, Krisciunas–Schaefer moonlight, twilight), a complete noise budget including scintillation and sky-subtraction noise, and per-night planning tracks with refraction-corrected airmass and parallactic angle. Every relation is documented, unit-tested, and inverted in closed form for target-S/N planning.

1 Motivation

Exposure time calculators are standard infrastructure at professional observatories, but each is bound to its facility, and the generic alternatives model idealized instruments. The communities producing a growing share of time-domain follow-up — variable-star and exoplanet photometrists, Be-star and nova spectroscopists on sub-metre telescopes — plan observations with rules of thumb, although their regime is the one where honest error budgets matter most: bright stars are scintillation-limited, urban skies are background-limited, and slitless gratings are sky-limited in a way slit intuition does not predict. SPETC (Spectro-Photometric Exposure Time Calculator) addresses exactly this regime. It descends from a 2007 Fortran chain, rewritten in Python on `astropy` [Astropy Collaboration, 2022], and runs fully offline as a desktop application, a headless batch tool, and an importable library.

2 Physics

Templates are absolutely calibrated CALSPEC/BPGS spectrophotometry [Bohlin et al., 2014], rescaled to the user’s magnitude in any reference filter through SVO-convention synthetic photometry [Rodrigo and Solano, 2020] with photon or energy weighting, and optionally transformed by a radial velocity and CCM89 reddening [Cardelli et al., 1989]. The sky model is position- and time-dependent: van Rhijn airglow attenuated along the slant path, zodiacal light as a function of ecliptic latitude and solar elongation, integrated starlight, Krisciunas and Schaefer [1991] moonlight with the lunar distance factor, and a twilight/daylight bridge on the geometric solar altitude. Airmass follows Pickering [2002] on refracted apparent altitudes; atmospheric dispersion follows Filippenko [1982] and feeds worst-case slit losses whenever the slit is not at the parallactic angle, which the planning track reports per time sample. Point-spread functions are Gaussian or Moffat with analytic encircled-energy and slit couplings (the

Moffat marginal is a Student- t distribution, so no profile is ever numerically integrated), with optional Kolmogorov seeing scaling in wavelength and airmass and a guiding-blur term. The noise budget is the CCD equation extended with scintillation [Young, 1967, Osborn et al., 2015], ADC quantization, dark, an extra-background term, and the Merline and Howell [1995] sky-subtraction factor; because the scintillation variance is linear in time, the target-S/N inversion and a stack planner (frames $\times$ sub-exposure, read-noise penalty, sky-limited frame length) remain closed-form even for scintillation-limited stars. Spectroscopic counts are exact per-resel integrals of the template — narrow-line fluxes are invariant under the chosen resolving power — and each output row carries the Cayrel [1988] equivalent-width uncertainty. Slitless mode implements Star Analyser physics: dispersion from groove density and grating-sensor distance, seeing-limited resolution, and the full-bandpass sky background that every pixel of a slitless spectrum receives, the dominant and often overlooked term. Figure 1 shows the resulting behaviour for a representative 358-mm configuration.

3 Implementation and availability

SPETC provides a Tkinter desktop interface with whole-night planning plots, a JSON-driven batch mode producing CSV and HTML summaries, tools to import CALSPEC/STScI FITS spectra as templates and to convert measured filter transmissions into fully annotated SVO-format profiles, CMOS gain-table support, SQM/Bortle sky input, differential-photometry precision in mmag, and a terrain-horizon generator based on the Copernicus GLO-30 elevation model. All physics is regression-tested (70+ tests), documented in a derivation-level manual, and was subjected to a full internal physics audit whose findings — including the slitless sky term and the resolution-invariance requirement — are implemented and tested. The code, documentation and audit are available under an open-source license in the project repository.

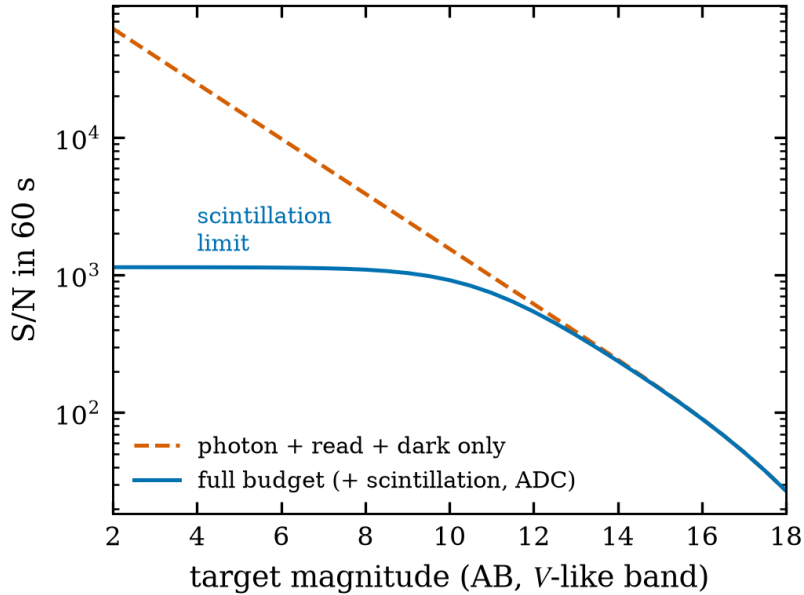


Figure 1: Broad-band signal-to-noise in 60 s versus target magnitude for a 358-mm telescope, computed by the released code. The full budget (solid) saturates at the scintillation ceiling near $S/N \simeq 1.2 \times 10^3$; a Poisson-only model (dashed) overestimates bright-star performance by more than an order of magnitude — the central planning error SPETC is designed to prevent.

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